

CO4_AT3_INDUSTRY PROBLEM SOLVING TASK

Scenario / Problem Context:

An educational platform conducts online quizzes for students across various subjects. Currently, quiz evaluation is manual, causing delays and errors in result processing. The platform requires a system to manage quizzes, evaluate answers automatically, and generate performance reports.

Task Description:

Develop a Python application using file handling and modules to manage quizzes and automate evaluation.

Sub-Tasks / Expected Deliverables:

- a) Define classes Quiz, Question, and StudentResponse.
- b) Store questions and answers in files.
- c) Implement evaluation logic to calculate scores.
- d) Generate result reports for each student.
- e) Handle invalid responses and missing answers.
- f) Maintain logs for quiz attempts.

ONLINE QUIZ AND EVALUATION SYSTEM

Python Program

models.py

This file contains the three required classes:

- Question
- Quiz
- StudentResponse

```
# models.py
```

```
class Question:
    def __init__(self, question_id, question_text, options,
correct_answer):
        self.question_id = question_id
        self.question_text = question_text
        self.options = options
        self.correct_answer = correct_answer
```

```
class StudentResponse:
    def __init__(self, student_name, question_id, answer):
        self.student_name = student_name
        self.question_id = question_id
        self.answer = answer
```

```
class Quiz:
    def __init__(self, title):
        self.title = title
        self.questions = []

    def add_question(self, question):
        self.questions.append(question)

    def display_questions(self):
        for question in self.questions:
            print(f"\nQ{question.question_id}.
{question.question_text}")

            for option in question.options:
                print(option)
```

quiz_manager.py

This module handles:

- Reading questions from files
- Taking student responses
- Validation
- Evaluation
- Report generation
- Logging

```
# quiz_manager.py
```

```
import os
from models import Question, StudentResponse
```

```
QUESTION_FILE = "questions.txt"
RESPONSE_FILE = "responses.txt"
LOG_FILE = "quiz_logs.txt"
REPORT_FOLDER = "reports"
```

```
def load_questions():
    questions = []

    try:
        with open(QUESTION_FILE, "r") as file:

            for line in file:
                line = line.strip()

                if not line:
                    continue

                data = line.split("|")

                question_id = int(data[0])
                question_text = data[1]

                options = [
```

```

        "A. " + data[2],
        "B. " + data[3],
        "C. " + data[4],
        "D. " + data[5]
    ]

    correct_answer = data[6].upper()

    question = Question(
        question_id,
        question_text,
        options,
        correct_answer
    )

    questions.append(question)

except FileNotFoundError:
    print("Error: questions.txt file not found.")

except Exception as e:
    print("Error while loading questions:", e)

return questions


def save_response(response):
    try:
        with open(RESPONSE_FILE, "a") as file:
            file.write(
                f"{response.student_name}|"
                f"{response.question_id}|"
                f"{response.answer}\n"
            )

    except Exception as e:
        print("Error saving response:", e)


def log_attempt(student_name, score, total):
    try:
        with open(LOG_FILE, "a") as file:

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        percentage = (score / total) * 100

        file.write(
            f"Student: {student_name} | "
            f"Score: {score}/{total} | "
            f"Percentage: {percentage:.2f}%\n"
        )

    except Exception as e:
        print("Error writing log:", e)

def evaluate_answer(student_answer, correct_answer):
    if student_answer == "":
        return "Missing"

    if student_answer not in ["A", "B", "C", "D"]:
        return "Invalid"

    if student_answer == correct_answer:
        return "Correct"

    return "Wrong"

def generate_report(student_name, results, score, total):
    if not os.path.exists(REPORT_FOLDER):
        os.makedirs(REPORT_FOLDER)

    percentage = (score / total) * 100

    if percentage >= 80:
        performance = "Excellent"
    elif percentage >= 60:
        performance = "Good"
    elif percentage >= 40:
        performance = "Average"
    else:
        performance = "Needs Improvement"

    filename = (
        REPORT_FOLDER + "/" +
        student_name.replace(" ", "_") +

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```

        "_report.txt"
    )

    try:
        with open(filename, "w") as file:

            file.write("=====\n")
            file.write("        ONLINE QUIZ RESULT REPORT\n")

file.write("=====\n\n")

            file.write(f"Student Name : {student_name}\n")
            file.write(f"Score       : {score}/{total}\n")
            file.write(f"Percentage  : {percentage:.2f}%\n")
            file.write(f"Performance : {performance}\n\n")

            file.write("Question-wise Result\n")
            file.write("-----\n")

            for result in results:
                file.write(
                    f"Q{result['question_id']} : "
                    f"Your Answer = {result['answer']} | "
                    f"Correct Answer = {result['correct']} | "
                    f"Result = {result['status']}\n"
                )

file.write("\n=====\n")

        return filename

    except Exception as e:
        print("Error generating report:", e)
        return None

def take_quiz(student_name, questions):

    responses = []
    results = []
    score = 0

```

```

print("\n=====")
print("                ONLINE QUIZ")
print("=====")

for question in questions:

    print(f"\nQ{question.question_id}.
{question.question_text}")

    for option in question.options:
        print(option)

    while True:

        answer = input("Enter your answer (A/B/C/D):
").strip().upper()

        # Missing answer
        if answer == "":
            print("No answer entered. Marked as Missing.")
            break

        # Invalid answer
        if answer not in ["A", "B", "C", "D"]:
            print("Invalid response! Please enter A, B, C or
D.")
            continue

        break

    response = StudentResponse(
        student_name,
        question.question_id,
        answer
    )

    responses.append(response)

    status = evaluate_answer(
        answer,
        question.correct_answer
    )

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```
    if status == "Correct":
        score += 1

    results.append({
        "question_id": question.question_id,
        "answer": answer if answer else "Not Answered",
        "correct": question.correct_answer,
        "status": status
    })

    save_response(response)

total = len(questions)

log_attempt(
    student_name,
    score,
    total
)

report = generate_report(
    student_name,
    results,
    score,
    total
)

return score, total, report
```


main.py

This is the main program.

```
# main.py

from models import Quiz
from quiz_manager import load_questions, take_quiz

def main():

    print("=====")
    print("      ONLINE QUIZ AND EVALUATION")
    print("=====")

    student_name = input("Enter student name: ").strip()

    if student_name == "":
        print("Student name cannot be empty.")
        return

    questions = load_questions()

    if len(questions) == 0:
        print("No questions available.")
        return

    quiz = Quiz("Python Programming Quiz")

    for question in questions:
        quiz.add_question(question)

    print(f"\nQuiz Title : {quiz.title}")
    print(f"Total Questions : {len(quiz.questions)}")

    input("\nPress Enter to start the quiz...")

    score, total, report = take_quiz(
        student_name,
        quiz.questions
    )

    percentage = (score / total) * 100
```

```

print("\n=====")
print("                QUIZ COMPLETED")
print("=====")

print(f"Student      : {student_name}")
print(f"Score        : {score}/{total}")
print(f"Percentage    : {percentage:.2f}%")

if percentage >= 80:
    print("Performance : Excellent")
elif percentage >= 60:
    print("Performance : Good")
elif percentage >= 40:
    print("Performance : Average")
else:
    print("Performance : Needs Improvement")

if report:
    print(f"\nReport generated successfully:")
    print(report)

print("\nThank you for taking the quiz!")

if __name__ == "__main__":
    main()

```

questions.txt

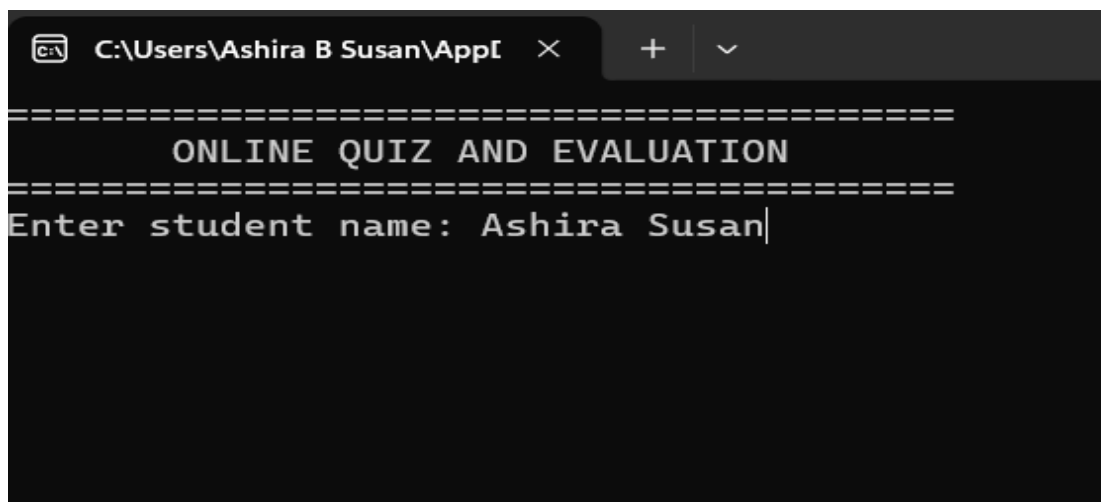
This is where the quiz questions and answers are stored.

The format is:

ID|Question|Option A|Option B|Option C|Option D|Correct Answer

- 1|Which keyword is used to define a function in Python?|func|define|def|function|C
- 2|Which data type is used to store True or False?|int|bool|string|float|B
- 3|Which symbol is used for comments in Python?|//|/* */|#|--|C
- 4|Which function is used to display output in Python?|input()|print()|display()|output()|B
- 5|Which keyword is used to create a class in Python?|object|class|struct|define|B

SAMPLE INPUT:



```
C:\Users\Ashira B Susan\AppData\Local\Microsoft\Windows\In...
```

```
=====
ONLINE QUIZ AND EVALUATION
=====
Enter student name: Ashira Susan|
```

```
C:\Users\Ashira B Susan\AppData\Local\Microsoft\Windows\In...
```

```
=====
      ONLINE QUIZ AND EVALUATION
=====
Enter student name: Ashira Susan

Quiz Title : Python Programming Quiz
Total Questions : 5

Press Enter to start the quiz...|
```

```
C:\Users\Ashira B Susan\AppData\Local\Microsoft\Windows\In...
```

```
=====
      ONLINE QUIZ AND EVALUATION
=====
Enter student name: Ashira Susan

Quiz Title : Python Programming Quiz
Total Questions : 5

Press Enter to start the quiz...

=====
      ONLINE QUIZ
=====

Q1. Which keyword is used to define a function in Python?
A. func
B. define
C. def
D. function
Enter your answer (A/B/C/D): C|
```

Press Enter to start the quiz...

```
=====
                ONLINE QUIZ
=====
```

Q1. Which keyword is used to define a function in Python?

- A. func
- B. define
- C. def
- D. function

Enter your answer (A/B/C/D): C

Q2. Which data type is used to store True or False?

- A. int
- B. bool
- C. string
- D. float

Enter your answer (A/B/C/D): B

Q3. Which symbol is used for comments in Python?

- A. //
- B. /* */
- C. #
- D. --

Enter your answer (A/B/C/D): C

Q4. Which function is used to display output in Python?

- A. input()
- B. print()
- C. display()
- D. output()

Enter your answer (A/B/C/D): B

Q5. Which keyword is used to create a class in Python?

- A. object
- B. class
- C. struct
- D. define

Enter your answer (A/B/C/D): A

OUTPUT REPORT:

```
File Edit View H1  B ...  🔍  👤  ⚙️

=====
      ONLINE QUIZ RESULT REPORT
=====

Student Name : Ashira Susan
Score       : 4/5
Percentage  : 80.00%
Performance : Excellent

Question-wise Result
-----
Q1 : Your Answer = C | Correct Answer = C | Result = Correct
Q2 : Your Answer = B | Correct Answer = B | Result = Correct
Q3 : Your Answer = C | Correct Answer = C | Result = Correct
Q4 : Your Answer = B | Correct Answer = B | Result = Correct
Q5 : Your Answer = A | Correct Answer = B | Result = Wrong

=====
```

RESPONSE TRACKER:

```
File Edit View H1  ▾

Ashira Susan|1|C
Ashira Susan|2|B
Ashira Susan|3|C
Ashira Susan|4|B
Ashira Susan|5|A
|
```